
虚空相位动力学 (The Phase Dynamics of the Void, PDV) 虚空相位宇宙学 (The Phase Cosmology of the Void)

一个基于三值共振虚空相位宇宙学框架

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提交日期: 2026 年 7 月

The Phase Dynamics of the Void, PDV

本文的思考源于一件生活中的小事：今年在给读初二的儿子辅导物理时，他的一连串追问，以及随后我与人工智能的对话，触发了我长久的沉思。自幼熟知的牛顿由苹果落地启发而发现了万有引力、伽利略因支持日心说而遭受迫害的故事，一直教导我们要敢于从简单现象中追问根本。面对自然界四种基本作用力强度相差悬殊这一事实，我不禁猜想，如果存在某种更深层的解释，能否像日心说取代地心说那样，大大简化那些看似不合理的现象，并揭示引力与核力之间的内在联系和底层原理？循着这一问题，我重新审视了现代物理学的图景：从“我们从哪里来”的追问引出大爆炸能量来源之谜，到背景辐射均匀度疑难所催生的暴胀理论，而大爆炸理论至今也没有解决能量来源问题；从望远镜观测的光线红移所暗示的宇宙膨胀，到星系旋转速度异常所指向的暗物质悬案，数十载搜寻仍一无所获；从广义相对论预言的黑洞——一个理论导致自身失效的逻辑矛盾，到宇宙加速膨胀所引出的暗能量方案，再到量子场论对真空能量的估算与观测之间那匪夷所思的 120 个数量级的落差；从量子力学中薛定谔的猫、多世界诠释与至今悬而未决的测量问题，到量子场论以极其复杂的数学模型高度拟合物理现实，标准模型那近二十个不可推导的自由参数、粒子质量的巨大层级与代际之谜、粒子寿命的悬殊差距，均无法得到根本性解释，反而背离了“越底层越简单”的科学直觉。此外，广义相对论与量子力学在奇点处同时失效的窘境、引力迟迟不能量子化的困境，以及宇宙中物质远多于反物质的源头，都指向同一个事实：物理学各自为政，诸多矛盾之处远未得到根本修补。就像日心说的提出那样，或许我们需要换一个角度，用更根本、更简单的第一性原理来重新理解世界——世界本不该那么复杂。正是在这种转换视角的尝试中，竟有豁然开朗之感。凭借物理直觉与反复深思，我形成了本篇猜想。

Translated by DeepSeek

The reflections in this essay originate from a small incident in daily life: while tutoring my son in physics when he was in his second year of junior high school this year, his series of persistent questions, followed by my subsequent dialogue with an artificial intelligence, triggered a prolonged period of contemplation. The stories I have known since childhood—of Newton being inspired by a falling apple to discover universal

gravitation, and of Galileo suffering persecution for supporting the heliocentric theory—have always taught us to dare to trace simple phenomena back to their fundamental roots. Confronted with the fact that the strengths of nature's four fundamental forces differ so enormously, I cannot help but wonder: if a deeper explanation exists, could it, much like the heliocentric model supplanting the geocentric one, greatly simplify those seemingly unreasonable phenomena and reveal an intrinsic connection and underlying principle between gravity and the nuclear forces?

Pursuing this question, I re-examined the landscape of modern physics: from the question “Where do we come from?” leading to the enigma of the Big Bang's energy source, to the inflation theory prompted by the puzzle of the uniformity of the cosmic microwave background—yet the Big Bang theory has still not resolved the issue of the energy source; from the redshift of light observed through telescopes implying an expanding universe, to the anomaly of galactic rotation speeds pointing to the unsolved mystery of dark matter, for which decades of searching have yielded nothing; from the black holes predicted by general relativity—a logical contradiction where a theory leads to its own breakdown—to the dark energy proposal brought forth by the discovery of the accelerating expansion of the universe, and on to the mind-boggling 120 orders of magnitude discrepancy between the quantum field theory estimate of vacuum energy and what is observed; from Schrödinger's cat, the many-worlds interpretation, and the still unresolved measurement problem in quantum mechanics, to the highly complex mathematical models of quantum field theory that fit physical reality with great precision, the nearly twenty non-derivable free parameters of the Standard Model, the enormous hierarchy and generation puzzle of particle masses, and the vast disparity in particle lifetimes—none of these can be fundamentally explained, and all instead deviate from the scientific intuition that “the more fundamental, the simpler.”

Furthermore, the predicament where both general relativity and quantum mechanics fail simultaneously at singularities, the persistent difficulty of quantizing gravity, and the origin of matter vastly outnumbering antimatter in the universe all point to the same fact: physics operates in disconnected domains, and its many contradictions are far from being fundamentally mended. Just as the heliocentric theory was proposed, perhaps we need to shift our perspective and use more fundamental, simpler first principles to re-understand the world—the world should not be this complicated. It was precisely in this attempt to change perspectives that a sudden sense of clarity emerged. Relying on physical intuition and repeated deep contemplation, I formed the conjecture presented in this essay.

黑洞等天体离我们太过遥远，不如就从身边的事情开始。生物学研究证实，人体由数量庞大的细胞构成，而这些细胞归根到底又是由原子组成的。细胞每天都在死亡与新

生，我们却对它们的生灭浑然无觉。粒子物理的研究同样揭示，大多数粒子的寿命都极为短暂，唯有电子和光子的寿命相对较长；更奇特的是，质子的稳定性高得超乎寻常——实验测得其寿命下限远远超过了宇宙的年龄。这一反常现象的背后，或许隐藏着某种更深层的东西。假如质子的真实情况恰好与我们观测到的相反——它其实和其他普通粒子一样，寿命极为短暂，只是因为我们自己就由质子构成，所以才无从觉察它的衰亡，这恰好对应了一句中国古诗：“不识庐山真面目，只缘身在此山中。”我们就以这个猜想为起点，展开后面的推演。

Translated by DeepSeek

Celestial bodies such as black holes are far too distant from us, so let us begin with matters closer to home. Biological research has confirmed that the human body is composed of an enormous number of cells, and these cells, in the final analysis, are made up of atoms. Every day, cells die and are born anew, yet we remain completely oblivious to their coming and going. Particle physics likewise reveals that the lifetimes of most particles are exceedingly short; only the electron and the photon have relatively long lifetimes. Even more peculiar is the extraordinary stability of the proton — its experimentally measured lower limit on lifetime far exceeds the age of the universe. Behind this anomaly, perhaps something deeper lies concealed. Suppose the real situation of the proton is exactly the opposite of what we observe — that, like other ordinary particles, its lifetime is in fact exceedingly short, only that we are utterly unable to perceive its decay because we ourselves are made of protons. This corresponds precisely to a classical Chinese poem: “Not knowing the true face of Mount Lu, simply because one is standing in its midst.” Let us take this conjecture as our starting point and proceed with the deduction that follows.

在量子场论中，真空并非空无一物，而是不断有虚粒子对瞬间产生又瞬间湮灭。这种“生—灭”是持续发生的，但它们无法被直接观测到，只能通过卡西米尔效应等可观测效应间接验证。这暗示了一个耐人寻味的可能性：质子是否也满足类似的生灭关系？在同一空间位置上瞬间湮灭又瞬间产生，这里不涉及同一个实粒子的“死亡与重生”，而是两个不同的全同粒子——因为生灭前后没有任何可分辨的差别，效果上等同于一次湮灭重生。为便于后续分析，我们将此定义为湮灭重生机制。据此，质子并非持续存在的实体，而是在极短的时间尺度上循环经历湮灭与全同再生。每一次湮灭释放出全部质量能量，随即在相同位置重新生成一个全同质子，将能量等量吸收，如此往复。由于生灭前后粒子全同、位置不变、能量收支严格平衡，任何由质子构成的观测者——包括我们自身——都天然地无法感知这一非连续过程。这恰是“不识庐山真面目，只缘身在此山中”的物理写照：质子呈现出的超常稳定性，并非粒子本身的持续存在，而是这一循环机制高度可靠的体现。由此来看，薛定谔那只既死又活的猫便有了真实的物理基础：若构成猫的每一个质子都处于生灭不息的循环之中，那么“生”与“死”的叠加态便可理解为生灭循环在不同相位上的宏观映射，猫的既死又活正是这一深层机制的自

然显现。

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Celestial bodies such as black holes are far too distant from us, so let us begin with matters closer to home. Biological research has confirmed that the human body is composed of an enormous number of cells, and these cells, in the final analysis, are made up of atoms. Every day, cells die and are born anew, yet we remain completely oblivious to their coming and going. Particle physics likewise reveals that the lifetimes of most particles are exceedingly short; only the electron and the photon have relatively long lifetimes. Even more peculiar is the extraordinary stability of the proton — its experimentally measured lower limit on lifetime far exceeds the age of the universe. Behind this anomaly, perhaps something deeper lies concealed. Suppose the real situation of the proton is exactly the opposite of what we observe — that, like other ordinary particles, its lifetime is in fact exceedingly short, only that we are utterly unable to perceive its decay because we ourselves are made of protons. This corresponds precisely to a classical Chinese poem: “Not knowing the true face of Mount Lu, simply because one is standing in its midst.” Let us take this conjecture as our starting point and proceed with the deduction that follows.

In quantum field theory, the vacuum is not truly empty; virtual particle pairs are continually popping into existence and annihilating in an instant. This “birth and death” is incessant, yet it cannot be observed directly — it can only be indirectly verified through observable effects such as the Casimir effect. This hints at a thought-provoking possibility: could it be that protons satisfy a similar relationship of birth and death? Instantly annihilating and instantly reappearing at the same spatial location — what is involved here is not the “death and rebirth” of the same real particle, but rather two distinct yet identical particles. Since there is no discernible difference before and after the process, the net effect is equivalent to an annihilation-rebirth. For the convenience of subsequent analysis, we define this as the Annihilation-Rebirth Mechanism. According to this mechanism, the proton is not a continuously existing entity; instead, it cyclically undergoes annihilation and identical regeneration on an extremely short timescale. In each cycle, it annihilates completely, releasing all its mass-energy, and immediately thereafter a fully identical proton is regenerated at the same location, absorbing an equal amount of energy, and so on in endless repetition. Because the particle before and after the cycle is identical, the position unchanged, and the energy budget strictly balanced, any observer composed of protons — ourselves included — is inherently incapable of perceiving this discontinuous process. This is precisely the physical embodiment of “Not knowing the true face of Mount Lu, simply because one is standing in its midst”: the extraordinary stability displayed by the proton is not a

manifestation of the particle's own continuous existence, but rather a reflection of the extraordinary reliability of this cyclic mechanism. Viewed in this light, Schrödinger's cat that is both dead and alive possesses a genuine physical foundation: if every proton constituting the cat is caught in an unending cycle of birth and death, then the superposition of "life" and "death" can be understood as a macroscopic projection of the birth-death cycle in different phases, and the cat's simultaneous life and death emerges as a natural manifestation of this deeper mechanism.

然而，这一推论的成立有一个不容回避的前提：我们何以确认质子的生灭循环不可观测？答案正藏在我们的感知边界之中。我们感知世界的通道，几乎被两种相互作用所垄断——远距依赖引力，近距依赖电磁，除此之外，别无他途。电磁观测依赖粒子在时间上的连续演化来传递信号，引力观测依赖质量在时空中的连续分布来产生效应。两者都以“探测对象的持续存在”为不言自明的工作前提。倘若质子本身并不连续，而是在生灭之间不断切换，那么我们的仪器——本身就是由质子构成的——便天然地被困在了同一个生灭节律之中，永远只能捕捉到循环完成之后的净结果，而非循环本身。换言之，不是质子不灭，而是我们没有能力看见它的灭。正是这一观测者局限，为湮灭重生机制提供了存在的空间，也使“存在即生灭”的猜想获得了认识论上的立足之地。

Translated by DeepSeek

However, the validity of this deduction rests on an unavoidable premise: how can we be certain that the proton's cycle of birth and death is unobservable? The answer lies precisely within the boundaries of our perception. Our channels for sensing the world are almost entirely monopolized by two interactions—gravitation at a distance, electromagnetism up close, and beyond these, there is no other path. Electromagnetic observation relies on the continuous evolution of particles through time to transmit signals; gravitational observation relies on the continuous distribution of mass in spacetime to produce effects. Both take the “persistent existence of the object under observation” as a tacit working premise. If protons themselves are not continuous, but instead switch incessantly between birth and death, then our instruments—themselves constructed from protons—are inherently trapped within the same rhythm of birth and death, forever capable only of capturing the net result after a cycle is complete, never the cycle itself. In other words, it is not that protons do not perish, but that we lack the capacity to witness their perishing. It is precisely this observer limitation that grants the Annihilation-Rebirth Mechanism its space to exist, and secures for the conjecture that “existence is birth-and-death” its foothold in epistemology.

若将质子的一个完整生死循环视为一个周期 (2π), 那么受限于电磁观测的固有属性, 我们永远无法捕捉湮灭发生的那一段相位, 而只能接收到重生完成之后的信号。由此, 我们被导向一个根本性的认知断裂: 电磁手段所呈现的宇宙, 只是完整宇

宙中被照亮的一道窄带。

下面引入本理论的一个重要概念。我们将宇宙依质子生命周期性划分为两个域：重生周期对应着我们日常所见的宇宙，我们称之为世界相；由于从质子重生到电磁作用建立之间存在着固有的时间差，电磁作用并非覆盖整个重生周期，电磁作用的有效范围我们称之为电磁相——它恰好对应于重生周期中电磁信号得以稳定传播的那段区间。电磁相是世界相的一个窗口，比世界相更窄，世界相的表象即由电磁相所呈现。为便于理解，可将每个相位类比为无线电中的频带：它有一个中心位置和一定的宽度——相位中心 AAA 决定了该相在周期中的定位，相位宽度 BBB 则标定了其覆盖的区间范围。世界相划定了一个观测者在生灭宇宙中理论上可触及的边界，而电磁相则标定了实际可捕获信号的那段窗口，后者嵌于前者之内，两者共同构成了认知的边界条件。

Translated by DeepSeek

If we regard a complete cycle of proton birth and death as one period (2π), then, constrained by the inherent properties of electromagnetic observation, we can never capture the phase segment in which annihilation occurs, and can only receive the signal after rebirth is complete. From this, we are led to a fundamental cognitive rupture: the universe presented by electromagnetic means is merely a narrow band illuminated within the complete universe.

Below, we introduce an important concept of this theory. We divide the universe into two domains according to the periodicity of the proton's life cycle: the rebirth cycle corresponds to the universe as we see it in daily life, which we call the World Phase; since there is an inherent time lag between the rebirth of a proton and the establishment of electromagnetic interaction, electromagnetic interaction does not cover the entire rebirth cycle. The effective range of electromagnetic interaction is what we call the Electromagnetic Phase — it corresponds exactly to the segment of the rebirth cycle in which electromagnetic signals can propagate stably. The Electromagnetic Phase is a window within the World Phase, narrower than the World Phase, and the appearance of the World Phase is presented by the Electromagnetic Phase. To facilitate understanding, each phase can be analogized to a frequency band in radio: it has a center position and a certain width — the phase center AAA determines the phase's position within the cycle, and the phase width BBB defines the range it covers. The World Phase delineates the theoretically accessible boundary for an observer in a universe of birth and death, while the Electromagnetic Phase marks the actual window in which signals can be captured; the latter is embedded within the former, and together they constitute the boundary conditions of cognition.

说到相位，我们很自然地想到交流电，那就从研究交流电开始吧。交流电的正半周和负半周都对应着能量的输出，都是可以做功的，而相位零点则能量为零。虽然在正负

半周上都可以输出能量，但从整体来看，交流电的正负半周是可以累加为零的。再从宏观尺度审视我们的世界相，有两个根本性的观测事实格外引人注目：其一，绝对零度不可达到。这不只是一个技术瓶颈，而更像是一条原理性的禁线。在我们的框架中，它恰好对应着交流电的相位零点——能量输出为零的那一瞬间。它标定的是电磁相窗口关闭的极限，是认知与世界相失去接触的那个边界。其二，宇宙中弥漫着 2.73K 的微波背景辐射。在我们的框架中，它对应的正是交流电中做功的那部分——它是宇宙物质能量总量在这个相位上的平均值。它并非来自某个具体事件的回声，而是世界相窗口内宇宙能量分布的有效值，就如同交流电的有效值表征其实际做功能力一样。绝对零度与 2.73K 背景辐射，由此构成了世界相的基本标度：前者是电磁相窗口闭合时的零能量节点，后者是窗口开启期间可观测能量的平均水平。我们身处的整个可观测宇宙，便落在这两个边界所框定的电磁窗口之内。

Translated by DeepSeek

Speaking of phases naturally brings alternating current to mind. So let us begin with the study of alternating current. In alternating current, both the positive half-cycle and the negative half-cycle correspond to energy output — both can perform work — while at the phase zero point, the energy is zero. Although energy can be output in both the positive and negative half-cycles, when viewed as a whole, the positive and negative half-cycles of alternating current can sum to zero.

When we examine our World Phase from a macroscopic scale, two fundamental observational facts stand out with particular prominence:

First, absolute zero is unattainable. This is not merely a technical bottleneck, but rather appears to be a prohibitive line of principle. In our framework, it corresponds exactly to the phase zero point of alternating current — that instant when energy output is zero. It marks the limit at which the Electromagnetic Phase window closes, the boundary where cognition loses contact with the World Phase.

Second, the universe is permeated with the 2.73K cosmic microwave background radiation. In our framework, it corresponds to the part of the AC cycle that performs work — it is the average value of the total material-energy content of the universe over this phase. It is not an echo of some specific event, but rather the effective value of the energy distribution of the universe within the window of the World Phase, just as the effective value of alternating current represents its actual capacity to perform work.

Thus, absolute zero and the 2.73K background radiation constitute the fundamental scale of the World Phase: the former is the zero-energy node when the Electromagnetic Phase window closes, the latter is the average level of observable

energy during the period when the window is open. The entire observable universe in which we reside lies within the electromagnetic window framed by these two boundaries.

至此，我们理应追问：世界相之外是什么？至少有一点可以确定——在周期序列上，必然存在着前一个世界相与后一个世界相。这里的前与后，仅表示它们在生灭周期中出现的时序。如果宇宙总共只有两个世界相，那么前一个和后一个便是同一个；若存在多个世界相，亦不改变推论的实质。为统一表述，我们以“前一个世界相”和“后一个世界相”来指称，由于后面的时间对称性，我们统称下一个相位。

交流电有正负半周，那么我们的宇宙是否也存在类似的正负结构？如果存在，它能否逻辑自治？让我们回到量子理论中的量子涨落：真空中可以凭空借出正反能量，瞬间产生又瞬间湮灭——这恰好对应着交流电的正半周与负半周。如果这一对应成立，我们便可自然地认为：宇宙中所有世界相的能量总和恒为零，各个世界相之间的差异，仅仅是相位上的能量分布不同，即功或功率的瞬时值不同。这个能量值，或许对应各相位的背景辐射温度，或许对应全周期上功率的平均值——具体的对应关系有待后续演算确定。

我们继续拆解交流电的特性。将直流电与交流电叠加，便得到带有偏置的交流电——它在一个恒定的基础能量之上振荡。这恰好对应着传统大爆炸理论的图景：宇宙需要一个巨大的初始能量作为“直流基底”，再叠加量子涨落这个“交流成分”，才能驱动宇宙的诞生并催生出星系结构的种子。

然而，引入世界相框架之后，这一图景便显得多余了。量子涨落本身——真空中正反能量的瞬间借出与归还——天然就是一个完整的交流周期，它本不需要一个直流偏置才能运作。大爆炸理论之所以需要一个初始能量，是因为它将量子涨落视为叠加在某个基底之上的扰动，而非宇宙能量运转的本体。但如果我们反过来看：量子涨落不是扰动，它就是宇宙能量运转的基本形式，那么直流偏置便不再必需。正半周的能量并非来自某个一次性注入的初始基底，而是来自前一个世界相在负半周中的“归还”——借与还首尾相衔，构成了一个自持的循环。

这里的关键在于，背景辐射所测量的，究竟是全周期的平均功率，还是本世界相内的能量密度？从背景辐射的监测手段来看，它完全依赖电磁观测，并未借助引力。这意味着它探测到的是本世界相内部的信息——即本世界相内的能量值，代表着世界相中物质的密度，对应的不是功率（即时值），而是功（累计和）。引入世界相理论之后，大爆炸理论所必需的那个基础能量便失去了存在的理由，暴胀解释也无需再登场。我们所做的，不过是将偏置直流换成了纯交流。

Translated by DeepSeek

At this point, we are compelled to ask: what lies beyond the World Phase? At least one thing is certain — in the periodic sequence, there must necessarily exist a preceding

World Phase and a succeeding World Phase. The terms “preceding” and “succeeding” here merely indicate their order of appearance within the birth-death cycle. If the universe has only two World Phases, then the preceding and the succeeding are one and the same; if there are multiple World Phases, the substance of the deduction remains unchanged. For the sake of uniform terminology, we refer to them as “the preceding World Phase” and “the succeeding World Phase,” and, given the time symmetry that follows, we collectively designate them as the next phase.

Alternating current has positive and negative half-cycles — does our universe likewise possess an analogous positive-negative structure? If so, can such a structure be logically self-consistent? Let us return to quantum fluctuations in quantum theory: in a vacuum, positive and negative energy can be momentarily borrowed out of nothing, instantly created and instantly annihilated — this corresponds exactly to the positive and negative half-cycles of alternating current. If this correspondence holds, we may naturally conclude that the sum of energy across all World Phases in the universe is identically zero, and that the differences among the various World Phases are merely differences in the distribution of energy over phase — that is, differences in the instantaneous value of work or power. This energy value might correspond to the background radiation temperature of each phase, or it might correspond to the average power over a full cycle — the exact correspondence awaits subsequent computational determination.

We continue to dissect the properties of alternating current. Superimposing a direct current upon an alternating current yields an AC with a bias — one that oscillates atop a constant baseline of energy. This corresponds precisely to the picture offered by traditional Big Bang theory: the universe requires an enormous initial energy to serve as the “DC baseline,” upon which quantum fluctuations — the “AC component” — are superimposed, in order to drive the birth of the universe and seed the structures of galaxies.

However, once the World Phase framework is introduced, this picture becomes superfluous. Quantum fluctuations themselves — the instantaneous borrowing and returning of positive and negative energy in the vacuum — inherently constitute a complete AC cycle, and they require no DC bias to operate. The reason Big Bang theory needs an initial energy is that it treats quantum fluctuations as perturbations superposed upon some baseline, rather than as the very substance of the universe's energy dynamics. But if we invert this perspective — if quantum fluctuations are not perturbations but the fundamental form in which the universe's energy operates — then a DC bias is no longer necessary. The energy of the positive half-cycle does not come from some one-time injection of an initial baseline; it comes from the

“repayment” made by the preceding World Phase during its negative half-cycle — borrowing and repayment linked end to end, forming a self-sustaining cycle.

The crucial question here is: does the measured background radiation represent the average power over a full cycle, or the energy density within our own World Phase? Judging from the means by which the background radiation is measured, it relies entirely on electromagnetic observation and makes no use of gravitation. This implies that what is being detected is information internal to our own World Phase — that is, the energy value within this World Phase, representing the density of matter in the World Phase, and corresponding not to power (an instantaneous value) but to work (a cumulative sum). With the introduction of the World Phase theory, the foundational energy required by Big Bang theory loses its reason for being, and the inflation hypothesis likewise need no longer be invoked. All we have done is replace a biased DC with a pure AC.

如此，绝对零度不可突破便有了理论依据：它对应着零相位的瞬时值，是所有粒子（波包）诞生的相位——处在这一位置的，都是刚刚重生的年轻粒子。这也将成为后续分析量子纠缠态的基础。此处引入粒子等同于波包的概念，在此不作展开。

还记得世界相吗？从本体层面来看，它直接对应着质子从出生到湮灭的整个过程。我们将质子存在的时间段定义为世界相。这里需要强调一点：并非质子生灭周期被动地与世界相同步，恰恰相反——先有质子，后有世界相。若质子的生灭周期发生漂移，世界相也随之漂移。世界相具有宽度 BBB，我们所见的粒子无一例外都落在这个宽度之内。这意味着什么？

一个耐人寻味的问题随之而来：光子和电子为何比其他粒子长寿得多？这两位究竟与众不同在何处？发光现象无处不在，光子的能量量子化为不同的能级，频率宽度极大，且没有静止质量；电子则不同，它拥有静止质量。从相位来看，唯一的不同在于它们偏离世界相中心相位的距离；从能量来看，则是波包大小的差异。粒子在相位上占据着不同的位置。那么，如果位于世界相边缘的粒子因重生时的相位偏移而跨越到另一个世界相，会发生什么？

这里我直接给出推论结果：光子，正是粒子跨世界相重生时进行能量结算的产物。重生的新粒子需要与新世界相的相位参数同步，由于两个世界相之间存在相位差，便触发了一次跳跃式的能量结算。结算能量的大小，取决于两个世界相的相位差以及粒子波包的大小。这正是光子与其他粒子特性迥异的根源——它并非实粒子，而是实粒子跨相位重生时能量结算所引发的激发，一次性释放出全部的相位差能量。这些能量的释放发生在两个世界相之间，不同步于任何单一世界相的周期，可以理解为在整个周期上均匀存在，是世界相扫过边缘粒子时所触发的结果。

那么电子又是什么？世界相边缘剧烈的能量跳跃，可以看作不同相位的波包——其中波包峰值相位与本世界相重合的那一部分，便表现为电子。电子的振荡周期小于世界

相的宽度，这正是电子云的形象化描述。若此描述对应于游离电子，那么在原子内部，同样存在着由类似能量跳变所激发的电子，这便是核内电子。电子的波峰相位位于世界相边缘附近，在重生涨落过程中可以与光子相互转换。原子内部的电子，是在质子等重粒子诞生之后才受激出现的，其数量与质子数一致。因此，核内电磁作用一定落在世界相之内，电磁相的宽度小于世界相。以上是我的强行推定，力求在逻辑上自洽。

Translated by DeepSeek

Thus, the unattainability of absolute zero gains a theoretical basis: it corresponds to the instantaneous value at phase zero, the phase at which all particles (wave packets) are born — those situated at this position are all newly reborn, young particles. This will also serve as the foundation for the subsequent analysis of quantum entanglement states. Here the concept that particles are equivalent to wave packets is introduced, but will not be elaborated further for now.

Do you still remember the World Phase? From the ontological perspective, it corresponds directly to the entire process of a proton from birth to annihilation. We define the time interval of a proton's existence as the World Phase. One point must be emphasized here: it is not that the proton's birth-death cycle passively synchronizes with the World Phase — quite the opposite — the proton comes first, and the World Phase follows. If the proton's birth-death cycle drifts, the World Phase drifts along with it. The World Phase has a width BBB , and every particle we observe falls within this width. What does this imply?

A thought-provoking question then arises: why do the photon and the electron live so much longer than other particles? What makes these two so exceptional? Luminescence is everywhere; the photon's energy is quantized into different energy levels, its frequency range is extremely broad, and it possesses no rest mass. The electron, on the other hand, does have rest mass. From the phase perspective, the only difference lies in their deviation from the central phase of the World Phase; from the energy perspective, it lies in the size of their wave packets. Particles occupy different positions in phase. Then, what happens if a particle near the edge of the World Phase, due to a phase shift during rebirth, crosses over into another World Phase?

Here I will directly state the deduced result: the photon is precisely the product of the energy settlement that occurs when a particle crosses World Phases during rebirth. The reborn new particle needs to synchronize its phase parameters with the new World Phase; because a phase difference exists between the two worlds, a discrete, jump-like energy settlement is triggered. The magnitude of the settled energy depends on the phase difference between the two worlds and the size of the particle's wave packet. This

is the root of the photon's distinct characteristics compared to other particles — it is not a real particle, but rather an excitation caused by the energy settlement when a real particle undergoes rebirth across phases, releasing the entire phase-difference energy at once. The release of this energy occurs between two World Phases, unsynchronized with the cycle of any single World Phase, and can be understood as being uniformly present across the entire cycle, a result triggered when the World Phase sweeps past edge particles.

So what, then, is the electron? The violent energy jumps at the edge of the World Phase can be viewed as wave packets at different phases — those wave packets whose peak phase coincides with our own World Phase manifest as electrons. The oscillation period of the electron is smaller than the width of the World Phase, and this is precisely a vivid description of the electron cloud. If this description applies to free electrons, then within the atom, electrons similarly excited by analogous energy jumps exist — these are the intra-atomic electrons. The peak phase of the electron lies near the edge of the World Phase, and during the fluctuations of rebirth it can interconvert with photons. Electrons within the atom appear only after the birth of heavy particles such as protons, and their number matches the number of protons. Therefore, the electromagnetic interaction within the nucleus must reside within the World Phase, and the width of the Electromagnetic Phase is smaller than that of the World Phase. The above is my forced postulation, striving for logical self-consistency.

基于以上分析，一个更深层的简化原则浮出水面。我们对质子、电子和光子的推论，均指向同一个方向：粒子的不同性质，并非源于它们本质上的差异，而仅仅源于它们在世界相中所处的相位位置不同。根据第一性原理，我们完全有理由将所有粒子视为全同粒子——它们本质上都是同一个实体（波包），只因在世界相周期中的相位坐标各异，才呈现出质量、电荷、寿命、稳定性等一系列看似截然不同的物理现象。

这一简化将使初期的推导难度大为降低：我们无需为每一种粒子单独设定属性，只需确定一个统一的波包模型，再通过它在世界相中的相位位置来导出其表观性质。质子的稳定，是因为它占据了世界相的中心相位；电子的双重性，是因为它处于世界相的边缘；光子的非实体性，是因为它本身不是粒子，而是粒子跨越世界相时能量结算的痕迹。其余粒子的复杂性质，亦可在这一框架中通过相位位置的差异化来逐步还原。后续，我们将在这一全同粒子的基础上，再进行差异化的推演。

Translated by DeepSeek

Based on the above analysis, a deeper simplifying principle emerges. Our deductions regarding the proton, electron, and photon all point in the same direction: the differing properties of particles do not stem from any essential differences among them, but solely from the different phase positions they occupy within the World Phase. According

to the first principle, we are fully justified in regarding all particles as identical particles — they are fundamentally the same entity (wave packet), and it is only because their phase coordinates within the World Phase cycle differ that they present a series of seemingly distinct physical phenomena such as mass, charge, lifetime, and stability.

This simplification greatly reduces the difficulty of the initial derivations: we no longer need to assign individual properties to each type of particle, but need only establish a unified wave packet model and then derive its apparent properties from its phase position within the World Phase. The stability of the proton arises because it occupies the central phase of the World Phase; the dual nature of the electron arises because it resides at the edge of the World Phase; the non-substantiality of the photon arises because it is not a particle itself, but the trace of the energy settlement when a particle crosses between World Phases. The complex properties of other particles can likewise be progressively reduced, within this framework, through the differentiation of their phase positions. In what follows, we will proceed with differentiated derivations on the foundation of these identical particles.

有感于量子涨落的思想与前文模型的推导，我们面临一个更深层的追问：全同粒子所诞生的“平台”本身，又该建立在怎样的基底之上？任何带有前提的设定——比如大爆炸的初始能量——都必然引发“前提从何而来”的无穷回溯。要使追问真正终止，基底本身必须是无前提的：它不能是“某种存在”，而只能是“无”。

由此，我们设想一个虚无的基底。它不是物理学意义上的真空，真空仍具有场和能量的结构，它是纯粹的无规定性——哲学上的“无”，数学上空集 ($\emptyset$) 所对应的本体。

但“无”如何才能产生“有”？这一跳跃曾被视为所有第一性原理推演中最脆弱的一环。然而，数学本身为我们提供了线索：空集 ($\emptyset$) 并非一无所有——它拥有一个非空的幂集 $\{\emptyset\}$ 。这意味着，即便在最纯粹的“无”之中，已然内蕴了“有”的可能性。虚无并非铁板一块的死寂，而是逻辑上天然允许着破缺——一种从“无”到“有”的跃迁，在数学上不仅是可能的，而且是必然蕴含于空集结构之中的。

基于此，我们假定：虚无基底发生了第一次破缺。这一破缺不需要原因，因为虚无不对任何事件设禁——它是绝对的无规定性，因而也不禁止自身的破缺。问“破缺为何发生”是一个无效的问题，因为虚无之中没有任何规则去阻止它发生，这正是虚空破缺逻辑上的合理性所在。

更进一步，正因为虚无基底在空间和时间上都没有任何先验结构，破缺的发生必然是非局域性的。它不发生于“某处”，因为“处”尚未存在；它不发生于“某时”，因为“时”亦未诞生。破缺的“发生”是弥漫性的、无优先位置的——它在任何可能的位置和时间上同时涌现。这意味着，从虚无中诞生的第一次破缺，天然覆盖了一切可能的坐标，天然具备均匀性与各向同性。

这恰好替代了暴涨理论的全部功能。暴涨理论之所以被引入，正是为了解释宇宙微波背景辐射在大尺度上的惊人均匀性——相距遥远的区域在没有因果接触的情况下，为何温度几乎完全相同。在标准宇宙学中，这需要一个极早期的指数式膨胀来“拉平”初始的不均匀。但在我们的框架中，均匀性不需要被制造出来——它是虚空破缺非局域性所天然携带的属性。第一次破缺在全局同时发生，它所借出的能量在一切位置上同步涌现，其后的一切演化均承袭了这一初始的均匀性。

由此，虚无基底的第一次虚空破缺——成为了一切的开端。它不是奇点，因为奇点意味着能量和时空的无限压缩；它是虚无的非局域性自发破缺，在全局同时借出正能量与负能量，分别对应粒子和反粒子的诞生。虚无在大部分时间里归于绝对的寂静，只在破缺的瞬间显现为“存在”。而我们的世界相，便是这一破缺所展开的存在窗口。

大爆炸理论所困扰的奇点问题、能量来源问题和暴涨问题，在此框架中一并消解：奇点不存在，因为基底从来不是奇点；能量不需要初始注入，因为能量来自虚空破缺的借出；暴涨不需要额外驱动，因为非局域性已天然确保了全局的均匀。一切，不过是从虚空中涌现的节律。

Translated by DeepSeek

Inspired by the idea of quantum fluctuations and the derivation of the preceding model, we are confronted with a deeper question: upon what kind of substrate should the platform that gives birth to identical particles itself be founded? Any premise-laden assumption—such as the initial energy of the Big Bang—inevitably leads to an infinite regression of the question “where did the premise come from?” To truly terminate this line of questioning, the substrate itself must be premise-free: it cannot be “some kind of existence,” but can only be “nothingness.”

Thus, we conceive of a void substrate. It is not the vacuum in the physical sense—a vacuum still possesses the structure of fields and energy—but pure indeterminacy: the “nothing” of philosophy, the entity corresponding to the empty set ($\emptyset$) in mathematics.

But how can “nothing” give rise to “something”? This leap has always been considered the most fragile link in all first-principle deductions. Yet mathematics itself provides us with a clue: the empty set ($\emptyset$) is not devoid of everything—it possesses a non-empty power set $\{\emptyset\}$. This means that even within the purest “nothing,” the possibility of “something” is already inherently contained. The void is not an ironclad stillness, but rather logically permits breaking—a transition from “nothing” to “something” that is not only possible mathematically, but is necessarily implied by the structure of the empty set itself.

Based on this, we postulate that the void substrate underwent a first breaking. This

breaking requires no cause, because the void imposes no prohibitions on any event—it is absolute indeterminacy, and therefore does not even forbid its own breaking. To ask “why did the breaking occur?” is an invalid question, for within the void there are no rules to prevent it from occurring. This is precisely the logical justification for the void breaking.

Furthermore, precisely because the void substrate has no a priori structure in space or time, the occurrence of breaking must be non-local. It does not happen “somewhere,” because “where” has yet to exist; it does not happen “at some time,” because “time” has yet to be born. The “occurrence” of breaking is diffusive and without privileged position—it simultaneously emerges at every possible location and time. This means that the first breaking born from the void naturally covers all possible coordinates, naturally possessing uniformity and isotropy.

This neatly replaces the entire function of inflation theory. Inflation theory was introduced precisely to explain the astonishing uniformity of the cosmic microwave background radiation on large scales—how regions far apart could have nearly identical temperatures without causal contact. In standard cosmology, this requires an extremely early exponential expansion to “smooth out” initial inhomogeneities. But in our framework, uniformity does not need to be manufactured—it is a property naturally carried by the non-locality of the void breaking. The first breaking occurred simultaneously everywhere; the energy it borrowed emerged synchronously at all positions, and all subsequent evolution inherits this primordial uniformity.

Thus, the first void breaking of the void substrate became the beginning of everything. It is not a singularity, for a singularity implies infinite compression of energy and spacetime; it is the spontaneous non-local breaking of the void, simultaneously borrowing positive and negative energy everywhere, corresponding respectively to the birth of particles and antiparticles. The void, for the vast majority of time, returns to absolute silence, manifesting as “existence” only in the instant of breaking. And our World Phase is precisely the window of existence unfolded by this breaking.

The problems that plagued the Big Bang theory—the singularity problem, the energy source problem, and the inflation problem—are all resolved within this framework: the singularity does not exist, because the substrate was never a singularity; energy does not require an initial injection, because energy comes from the borrowing of the void breaking; inflation does not need an additional driver, because non-locality naturally guarantees global uniformity. Everything is merely a rhythm emerging from nothingness.

为了简化计算并进行形象化表达，我们引入冲激函数来描述原初破缺，并使用序列

$\{+1,0,-1,0\}\{+1,0,-1,0\}\{+1,0,-1,0\}$ 来表示冲激发生的过程。由此，这一过程可被视作一种波——一种原初的随机振荡。其中， $+1+1+1$ 和 $-1-1-1$ 仅表示能量的借出方向（正能与负能），不表示能量的大小。该序列满足如下约束：

$\sum \sum \text{所有冲激} = 0 = 0$ ，且冲激必须满足 $\{+1,0,-1,0\}\{+1,0,-1,0\}\{+1,0,-1,0\}$ 或其循环置换的相位关系。

这一过程的物理图景如下：虚空在某一点借出正能量，便在对应位置留下一个负能量的空洞——这个空洞，就是空间；同理，虚空在某一点借出负能量，便在对应位置留下一个正能量的空洞——这个空洞，同样是空间。物质与空间乃一体两面：一个表示能量本身，一个表示能量的外表面。

由于破缺的随机本性，振荡的能量大小等性质极为随机。这些振荡互相影响、互相作用，但总体上保持能量守恒。正是这些相互作用，驱使破缺从最初的混沌逐步走向有序与稳定——这一过程类似于共振摆的同步现象。为便于表述，我将这一演化过程划分为三个时期：

混沌期：新生破缺的数量远大于共振一致的数量，此起彼伏，无法区分“这里”与“那里”，时间上毫无周期性，混沌一片。

共振期：共振一致的数量逐渐超过新生破缺的数量，振荡幅度与周期逐步趋于稳定，有序的节律开始浮现。

稳定期：共振一致的数量与新生破缺的数量达到稳定比例，一切基本的物理常数在此阶段定形。

破缺能量的涨落在空间上的发生，表现为空间上的能量守恒；破缺能量的涨落在周期上的先后发生，表现为时间上的能量守恒。这一守恒可同时体现于空间位置与振荡周期之上——例如，在某一片区域内、某一段周期中，能量保持守恒。这恰好体现了空间与时间的对称性。

我们将发生破缺（涨落）的位置定义为时空晶格。稳定期后期，一切稳定的物理参数得以确立——振荡周期对应于普朗克时间，传播速度对应于光速，距离对应于普朗克长度。共振期决定了时间与空间离散化的性质，这一时期的涨落与混沌期的全同性随机截然不同，呈现出清晰的规律性与位置确定性。这，正是空间感与时间感的由来——时间与空间，至此得以区分，至此诞生。

空间上的涨落相对直观，可直接对应于量子场论中的真空涨落：它同时发生于空间的两点。若在空间区域内，涨落首尾相接形成环状结构，便会构成稳定的空间漩涡。这一漩涡犹如空间中的暗桩，与时间上的涨落相互作用，对后者形成阻滞——这一机制，恰与希格斯场的机制相通，赋予粒子以质量。空间上的振荡序列可表示为 $\{+1,0\}\{+1,0\}\{+1,0\}$ 和 $\{-1,0\}\{-1,0\}\{-1,0\}$ ，或 $\{-1,0\}\{-1,0\}\{-1,0\}$ 和 $\{+1,0\}\{+1,0\}\{+1,0\}$ 。

时间上的振荡则有所不同，其序列可表示为 $\{+1,0,-1,0\}$ $\{+1, 0, -1, 0\}$ $\{+1,0,-1,0\}$ 或 $\{-1,0,+1,0\}$ $\{-1, 0, +1, 0\}$ $\{-1,0,+1,0\}$ ——这个，才是形成我们宇宙万物的根基。为便于沟通，我们将时间振荡定义为三值谐振（共振）。

为了维持时空晶格的离散性，虚空上每一点的涨落严格归属于该点自身——涨落所借出的能量只能就地湮灭，不能连续位移。这是时空晶格最基本的特性：空间上能量或物质的移动，必须遵循湮灭重生的规律，而非平滑的连续滑移。

由此，物质的运动不再是经典意义上的轨迹推移，而是一个离散的状态更新过程：每一个时空晶格在下一刻的状态，由其自身的前一状态与周围邻居的前一状态共同决定。这便构成了一个局域的、离散的动力学规则——物质在空间中的传播，本质上是一连串有序的湮灭与重生在相邻晶格之间的接力。

这恰好赋予惠勒那句名言以全新的表述。原句为：“物质告诉空间如何弯曲，空间告诉物质如何运动。”在世界相理论的框架中，我们将其改写为：

物质告诉空间如何重生，空间告诉物质如何运动。

前者对应物质对空间的作用——物质的存在改变了局域晶格上湮灭重生的边界条件；后者对应空间对物质的反作用——晶格之间状态更新的规则，决定了物质重生的位置与次序，从而呈现出我们所见到的运动。空间不再是连续弯曲的几何舞台，而是一张离散的、不断自我更新的时空晶格之网。

Translated by DeepSeek

To simplify calculations and enable a vivid representation, we introduce the impulse function to describe the primordial breaking, using the sequence $\{+1,0,-1,0\}$ $\{+1, 0, -1, 0\}$ $\{+1,0,-1,0\}$ to represent the process of impulse occurrence. Thus, this process can be viewed as a wave — a primordial random oscillation. Here, $+1+1+1$ and $-1-1-1$ only indicate the direction of energy borrowing (positive and negative energy), not the magnitude of the energy. The sequence satisfies the following constraint:

$\sum \sum \text{all impulses} = 0 = 0$, and the impulses must satisfy the phase relationship of $\{+1,0,-1,0\}$ $\{+1, 0, -1, 0\}$ $\{+1,0,-1,0\}$ or its cyclic permutations.

The physical picture of this process is as follows: when the void borrows positive energy at a point, it leaves behind a hole of negative energy at the corresponding location — this hole is space; similarly, when the void borrows negative energy at a point, it leaves behind a hole of positive energy at the corresponding location — this hole, too, is space. Matter and space are two sides of a single coin: one represents the energy itself, the other represents the exterior surface of the energy.

Due to the random nature of the breaking, the magnitude of the oscillation energy and other properties are extremely random. These oscillations influence and interact with

one another, but overall maintain energy conservation. It is precisely these interactions that drive the breaking from initial chaos toward order and stability — a process analogous to the synchronization of coupled pendulums. For ease of description, I divide this evolutionary process into three periods:

Chaotic Period: The number of nascent breakings far exceeds the number of those in resonant consensus, rising and falling ceaselessly, with no way to distinguish “here” from “there,” utterly aperiodic in time — a chaos without form.

Resonance Period: The number of those in resonant consensus gradually surpasses the number of nascent breakings; the oscillation amplitude and period progressively stabilize, and an orderly rhythm begins to emerge.

Stable Period: The number of those in resonant consensus and the number of nascent breakings reach a stable ratio; all fundamental physical constants take shape during this stage.

The manifestation of the breaking energy's fluctuations in space appears as the conservation of energy in space; the successive occurrence of the breaking energy's fluctuations across cycles appears as the conservation of energy in time. This conservation can manifest simultaneously over spatial positions and oscillation cycles — for example, within a certain region and over a certain period, energy remains conserved. This perfectly embodies the symmetry of space and time.

We define the location where a breaking (fluctuation) occurs as a **spacetime lattice**. In the late Stable Period, all stable physical parameters are established — the oscillation period corresponds to the Planck time, the propagation speed corresponds to the speed of light, and the distance corresponds to the Planck length. The Resonance Period determines the discrete nature of time and space; the fluctuations of this period are entirely different from the universal randomness of the Chaotic Period, displaying clear regularity and positional definiteness. This is the origin of the sense of space and the sense of time — time and space, at this point, can be distinguished and are thus born.

Spatial fluctuations are relatively intuitive and can directly correspond to the vacuum fluctuations in quantum field theory: they occur simultaneously at two points in space. If, within a region of space, fluctuations connect head to tail to form a ring-like structure, a stable spatial vortex is constituted. This vortex is like a hidden stake in space, interacting with temporal fluctuations and exerting a drag upon the latter — this mechanism is analogous to that of the Higgs field, endowing particles with mass. The oscillation sequences in space can be expressed as $\{+1,0\}\{+1,0\}\{+1,0\}$ and $\{-1,0\}\{-1,0\}\{-1,0\}$, or $\{-1,0\}\{-1,0\}\{-1,0\}$ and $\{+1,0\}\{+1,0\}\{+1,0\}$.

Temporal oscillation, however, is somewhat different; its sequence can be expressed as $\{+1,0,-1,0\}$ $\{+1, 0, -1, 0\}$ $\{+1,0,-1,0\}$ or $\{-1,0,+1,0\}$ $\{-1, 0, +1, 0\}$ $\{-1,0,+1,0\}$ — this is the very foundation that forms all things in our universe. For ease of communication, we define temporal oscillation as **Three-Valued Resonance**.

To maintain the discreteness of the spacetime lattice, every fluctuation in the void strictly belongs to that point itself — the energy borrowed by a fluctuation can only annihilate locally and cannot undergo continuous displacement. This is the most fundamental property of the spacetime lattice: the movement of energy or matter in space must obey the law of annihilation and rebirth, rather than smooth, continuous sliding.

Thus, the motion of matter is no longer the trajectory push of classical depiction, but a discrete state-updating process: the state of each spacetime lattice in the next moment is jointly determined by its own previous state and the previous states of its neighbors. This constitutes a local, discrete dynamical rule — the propagation of matter in space is essentially a relay of ordered annihilation and rebirth between adjacent lattices.

This perfectly endows John Wheeler's famous dictum with a brand-new formulation. The original reads: "Matter tells space how to curve; space tells matter how to move." Within the framework of the World Phase theory, we rewrite it as:

Matter tells space how to rebirth; space tells matter how to move.

The former corresponds to the action of matter on space — the existence of matter alters the boundary conditions of annihilation and rebirth on the local lattice; the latter corresponds to the reaction of space on matter — the state-updating rules between lattices determine the location and sequence of matter's rebirth, thereby giving rise to the motion we observe. Space is no longer a continuously curved geometric stage, but a discrete, ceaselessly self-renewing web of spacetime lattices.

层次	内容	状态
终极基底	虚无=空集($\emptyset$), 虚空破缺, $\{+1,0,-1,0\}$ $\{+1,0,-1,0\}$ 冲激序列	✓
本体论	物质与空间一体两面, 能量借出与空洞	✓
时空起	混沌→共振→稳定三阶段, 时空晶格离散化	✓

层次	内容	状态
源		
物理学常数	普朗克尺度参数在稳定期涌现	✓
质量起源	空间漩涡=暗桩，阻滞时间涨落	✓
离散动力学	湮灭重生接力，物质告诉空间如何重生，空间告诉物质如何运动	✓
粒子本质	全同粒子=波包，相位位置决定表观性质	✓
光子/电子	光子=跨相能量结算，电子=世界相边缘波包峰值	✓
三值谐振	时间振荡的数学表达	✓

至此，我们已熬制出一锅足以承载量子力学的底汤——它以虚空破缺为炉灶，以 $\{+1,0,-1,0\}\{+1,0,-1,0\}\{+1,0,-1,0\}$ 三值冲激序列为薪火，以离散时空晶格为骨架，以湮灭重生机制为律动。这锅底汤的滋味，正在于它使那些长久以来被视为公设的量子力学教条，变成了从更底层原理中自然析出的结晶。

下一步，便是将这锅底汤交付给量子力学进行修补——在离散晶格的舞台上，重新推导波函数、算符与演化方程，将测量坍缩还原为电磁相窗口内的晶格共振同步，将不确定性归结为晶格最小尺度的内禀属性。若此猜想最终成立，量子力学的计算框架将得以极大简化：希尔伯特空间不再是强加于自然之上的抽象数学外衣，而是世界相内相位配置的忠实映射；重整化无需人为截断，因为晶格本身就是最自然的紫外屏障；测量问题不再是幽灵般的坍缩，而是一场在湮灭重生节律中被精准“抓拍”的快照。

而那只有名的猫，也将不再是被囚禁在思想实验中的荒诞幽灵。既死又活的叠加态，并非逻辑的悖谬，而是生灭循环在不同相位上同时存在的宏观映照——猫的确同时活着与死去，只不过这两个状态分属于世界相内微微错开的相位切片。它并非既死又活

的矛盾体，而是跨越生灭节律的全息投影。薛定谔的猫，确实存在。

Translated by DeepSeek

Schrödinger's cat truly exists. Its simultaneous life and death is not a logical paradox, but a macroscopic reflection of the birth-death cycle coexisting across different phases—the cat does indeed live and die at once, only these two states belong to slightly staggered phase slices within the World Phase.

Yet a cat both dead and alive has already led us to a deeper question: if all things are but phases, what is it that connects these phases to one another? What causes one particle to tend toward another, and what causes them to move apart at some boundary? Between the two states of the cat, does there also exist some kind of “force” pulling at them?

At this point, we have finally arrived at the ultimate question that has remained unresolved since Newton—the nature of force.

We have already understood that the void breaks to produce space, and space converges to restore the void—this is the most fundamental contradiction between void and space. Yet breaking is not a chaotic, patternless fluctuation, nor is convergence an undifferentiated dissolution of structure. Between breaking and convergence, there exists a deeper constraint: symmetry.

The void is absolutely symmetric. Before breaking occurs, the void has no up or down, no front or back, no positive or negative, no before or after—it is a complete, undifferentiated symmetric entity. The occurrence of breaking is, in essence, the first breach of this absolute symmetry: borrowing positive energy leaves behind a hole of negative energy at the same instant, in another aspect of the same location. The separation of positive and negative is the first step of symmetry moving from “non-differentiation” toward “differentiation.”

Yet breaking's breach of symmetry is never total. In every breaking, the void enforces an inviolable law: global symmetry must be conserved. The borrowing of positive energy must be recorded, somewhere and at some time, in the form of negative energy; for every difference created, a corresponding reverse difference must balance the ledger. This law is the most fundamental expression of energy conservation at the level of the void—energy conservation is not a law that the universe evolved later, but an accounting principle forcibly executed by the void's symmetry in the process of breaking.

Thus, what breaking produces is never an isolated energy cluster, but paired, symmetrically correlated energy pairs—positive and negative, borrowed and returned,

here and there. The marks these energy pairs leave upon the spacetime lattice constitute the entire content of space.

Space convergence, then, is the reverse process by which symmetry moves from “broken” toward “restored.” Since breaking creates difference, space convergence exists precisely to eliminate difference and restore symmetry. This is the deep reason why space possesses the inherent tendency to “converge and restore the void”: space attempts to eliminate itself not because some external force compels it, but because the symmetry of the void has never truly relinquished its rule—each breaking is only a temporary, local rebellion, and symmetry, as the foundational law of the void, is always attempting to reclaim its lost territory through various convergent pathways.

Thus, the two pathways of space convergence—in-phase merging and anti-phase annihilation—receive a deeper symmetry-based explanation:

In-phase merging is the attempt of those identities in the void that have been “separated too close” to reassemble. Two energy clusters of the same sign, though each has breached the symmetry of the void, have their difference pointing in the same direction (both positive or both negative), so the symmetry between them has not been severely damaged. The convergence strategy of space is: let them merge, reduce the total surface area, narrow the scope of difference, and approach the void through topological contraction. This is a gentle restoration of symmetry—existence need not be eliminated, only made more compact.

Anti-phase annihilation is the final encounter of those opposites in the void “destined to cancel each other out.” Two energy clusters of opposite sign, one borrowing positive and one borrowing negative—the difference between them is a difference of polarity, a complete difference. Here, space has no room for compromise—merging is meaningless, contraction is out of the question, and the only way to restore symmetry is to let them directly offset and zero out. This is a thorough restoration of symmetry—existence is utterly erased in that instant, and the void recovers its undifferentiated ground state.

From this, we understand why void breaking must occur in pairs, why space convergence has precisely these two pathways and no others, and why attraction and repulsion—the two most fundamental mechanical effects—are ultimately rooted in the same single source: the breaking and restoration of the void’s symmetry. Force is nothing other than that irresistible push and pull imposed by symmetry upon spatial structures as it carries out its self-restoration.

The Nature of Force: The Macroscopic Manifestation of Phase Correction Driven by Space Convergence

Within the Void Phase Dynamics framework, force is no longer a mysterious, externally assigned property, but an intrinsic behavior of the void in maintaining energy conservation and space convergence. All forces—whether gravity, electromagnetism, the strong force, or the weak force—are fundamentally different manifestations of the same single mechanism: when a phase deviation arises between two oscillations, the void spontaneously initiates a correction, attempting to pull the phases back into synchrony. This correction process is what we perceive at the macroscopic level as “force.” The void possesses the inherent tendency to break and produce space, and once space is produced, it possesses the inherent tendency to converge and restore the void. Force is precisely the pressure that space exerts upon matter as it attempts to eliminate itself and return to the void's ground state—it is the struggle of “existence” attempting to become “non-existence” again. The void does not permit energy to appear or disappear without accounting. When the oscillation phases of two particles (wave packets) are inconsistent, there exists between them a region of energy imbalance—a “deficit” or “surplus” has appeared in the energy ledger of the void. More importantly, this phase inconsistency also signifies a non-optimal configuration of space—the surface area of the structure has not been compressed to its minimum, and the restoration of the void's ground state has been obstructed. In order to balance the accounts and compress space, the void drives these two particles closer together or farther apart, until their phases resynchronize and the surface area of space is minimized. This driving process is the origin of force.

Two Correction Modes: Attraction and Repulsion

Phase correction follows a simple rule, determined by the magnitude of the phase difference $\Delta\theta$ between two oscillations:

Mode One: Attraction (for impulse functions, $\Delta\theta < \pi$; for sinusoidal and Gaussian wave packets, $\Delta\theta < \pi/2$) When the phase difference is less than π , the void judges that these two oscillations “ought to be synchronized.” The energy deficit between them needs to be filled, and the way to fill it is to bring them closer together—the closer they are, the smaller the phase difference, and the nearer the energy approaches equilibrium. From the perspective of space convergence, the merging of two same-sign energy clusters (both positive or both negative) belongs to the isoperimetric contraction process of in-phase merging—the total volume after merging is conserved, but the total surface area decreases, and the spatial structure degrades toward the void's ground state. This process manifests as **attraction**. In the macroscopic world, since the vast majority of phase deviations in natural states are less than π , attraction is the most common manifestation of force in the universe. Gravity is precisely the embodiment of this mode at the macroscopic scale.

Mode Two: Repulsion (for impulse functions, $\Delta\theta > \pi$; for sinusoidal and Gaussian wave packets, $\Delta\theta > \pi/2$) When the

phase difference exceeds π , the situation flips. At this point, the void judges that these two oscillations “cannot be converged through merging”—the energy barrier that must be crossed to pull them back into synchrony exceeds the energy required to let them separate completely. The void's choice is: push them over, into the next synchronization cycle. The correction mechanism switches from “in-phase merging” to “driving toward anti-phase annihilation,” manifesting at the macroscopic level as **repulsion**. This is not another kind of force standing in opposition to attraction, but rather another branch of the same single space convergence process—when space cannot eliminate itself through merging, it creates the conditions for anti-phase annihilation through separation, zeroing out completely through annihilation. Push and pull alike are both for the sake of returning existence to non-existence. **The**

Annihilation Effect at Extreme Phase Difference When $\Delta\theta = \pi$, that is, when two oscillations are exactly in anti-phase, space convergence enters its most efficient mode. At this moment, the crest of the positive half-cycle meets exactly the trough of the negative half-cycle, and the two cancel each other out directly—one borrowing positive energy, the other borrowing negative energy, and the energy ledger of the void at that location is instantly zeroed out. This manifests as **annihilation**: the two particles disappear simultaneously at the point of encounter, their energy completing a cross-phase settlement in the form of photons, and the local hole vanishes completely, restored to the void's ground state. Annihilation is not destruction, but the most thorough execution of space convergence—no merging needed, no separation needed, just a direct offset to zero. In that instant, existence perfectly becomes non-existence again. This mechanism is key to understanding why the strong force is so powerful at extremely short distances: the phase difference between quarks is strictly locked within an extremely narrow range ($\Delta\theta \approx 0$), and any deviation triggers the most intense convergence pressure from space—space frantically attempts to pull the quarks back together to merge, to reduce surface area and restore the void. When quarks attempt to separate to a distance sufficient to produce new particles, space realizes that merging is no longer possible, and instead chooses to create new quark-antiquark pairs as a way of manufacturing the conditions for anti-phase, solving the problem thoroughly through annihilation—this is precisely the phase-based explanation of quark confinement. **The Transmission of Force: A**

Relay of Phase Settlement Signals and Space Convergence Pressure How is force transmitted from one particle to another? In Void Phase Dynamics, the transmission of force is not the exchange of some mysterious particle, but rather **a relay propagation of phase settlement signals and space convergence pressure across the spacetime lattice**. When the oscillation phase of a particle changes, it generates a phase disturbance on the lattice, simultaneously creating a local region of energy imbalance in

space—this means that the surface area of space has not been compressed to its minimum, and the convergence pressure has not yet been released. This disturbance and pressure propagate at the speed of light to adjacent lattices; each lattice, upon receiving the information, updates its own phase state, transmits the convergence pressure onward, and continues the propagation downstream. When the information reaches another particle, the latter adjusts its own oscillation according to the received phase information—if the phase difference between the two falls within the mergeable interval, the space convergence pressure drives them closer, executing in-phase merging; if within the unmergeable interval, it drives them apart, manufacturing the conditions for anti-phase annihilation. This entire chain of disturbance propagation, pressure transmission, and convergence execution constitutes what we call “the transmission of force.” Photons, gluons, and other gauge bosons in this picture are not “force-carrying particles,” but rather **energy packets triggered and released under different conditions by the phase settlement signals and space convergence pressure**—they are byproducts produced when space carries out its convergence operations, not the carriers of force.

The Special Character of Gravity Gravity occupies a special position among the four forces: its range of action is infinite, yet its strength is extremely weak. In Void Phase Dynamics, this originates from the fact that the frequency difference Δf corresponding to gravity approaches zero. Gravity is the macroscopic accumulation of the phase gradient produced by the collective oscillation of a vast number of identical particles (wave packets) in a macroscopic body—the phase correction contribution of each individual particle is negligible, and the space convergence pressure is extremely dilute, but when countless particles gather together, their collective effect manifests as gravity. The infinite range of gravity implies that its corresponding Δf is nearly zero—at the particle level, this means that gravity is not a resonance phenomenon of some specific particle, but rather the foundational phase background produced by the very existence of all particles within the World Phase. In other words, **gravity is the shadow of “existence” itself**: as long as matter exists, space has not completely converged back to the void, and the residual space continues to exert a faint convergence pressure. As long as matter exists, there is a phase gradient; as long as there is a phase gradient, there is a convergence of space that has not yet been completed—and this, precisely, is gravity. This is perhaps the deep reason why gravity has stubbornly resisted unification with the other three forces and has also resisted quantization: it is not a particular, special kind of force, but rather the global residual of the void's phase correction and space convergence mechanisms in the limit where $\Delta f \rightarrow 0$. Since phase difference can simultaneously give rise to both attractive and repulsive forces, there is a special feature of gravity: at close range it manifests as attraction, but once the phase exceeds a wavelength, it

manifests as repulsion. This is the cause of cosmic expansion—yet in reality, the universe is not expanding, because the universe is infinite in extent, and energy cannot undergo continuous displacement across lattices. The expansion we see is merely an illusion caused by phase drift.

Let the explanation of force end here. This has been but a brief account of the case of the impulse function; the principle is too simple, and the descriptions for various actual wave cases are too complex. Readers may grasp the principle for themselves; the computational part is left to the mathematicians. Below are appended some reinterpretations of various phenomena:

Reinterpretations of Several Cosmological Puzzles

Based on the World Phase framework, we can offer a unified explanation for puzzles such as dark matter, dark energy, black holes, redshift, cosmic expansion, and time dilation:

Dark matter corresponds to matter in the same-phase World Phase that lies outside the Electromagnetic Phase. It resides in the same phase cycle as our World Phase, and therefore participates in gravitational interactions, but because the phase position of its wave packets does not fall within the Electromagnetic Phase window, it cannot be observed through electromagnetic means. This is the fundamental reason why dark matter manifests gravitational effects yet remains invisible.

Dark energy corresponds to anti-phase matter—that is, antimatter, the part corresponding to the negative half-cycle in the alternating current model. The total positive and negative matter of the universe is strictly symmetric, and the sum of energy is identically zero. The “matter-antimatter asymmetry” we observe is not an imbalance of the universe as a whole, but simply arises because our World Phase happens to reside in the positive half-cycle, while antimatter exists in the phase staggered apart from it, the two never meeting within the Electromagnetic Phase window.

Black holes are not celestial bodies in the traditional sense, still less do they harbor singularities. They are regions formed after a severe phase desynchronization; the lattice update rhythm of these regions has completely decoupled from that of our World Phase, in essence corresponding to another World Phase. A black hole can be understood as a “parallel world” in the spatial domain—it still rests upon the spacetime lattice, but the phase of its annihilation and rebirth is no longer synchronized with our Electromagnetic Phase window, and therefore we cannot peer into its interior through electromagnetic means.

Redshift has two sources. First, photons in the course of propagation traverse multiple

World Phases, and each crossing of a phase boundary involves an energy settlement, accumulating a phase deviation—this is a spatial phase error accumulation effect. Second, gravity itself has a wavelength; when the spatial scale exceeds the gravitational wavelength, the phase of gravity undergoes reversal, manifesting as a repulsive effect, causing additional losses to the photon's energy and likewise producing redshift.

Cosmic expansion is, within this framework, an unnecessary and meaningless concept. The motion of matter inherently involves no continuous displacement, only a relay of annihilation and rebirth on the discrete lattice. The universe is infinite in extent, and discussing the expansion of an infinite space lacks meaning in itself. The observed recession of galaxies attributed to “cosmic expansion” is actually the second mechanism of redshift—the repulsive effect produced by the phase reversal of gravity when spatial scales exceed the gravitational wavelength, making distant celestial bodies appear to be accelerating away. This is merely a manifestation of gravitational phase in the course of transmission, not an expansion of space itself.

Time dilation—the slowing of time's passage for rapidly moving objects—follows the same principle as redshift. High-speed motion means that an object traverses more lattices per unit time, and its rhythm of annihilation and rebirth is disturbed, producing an effect akin to “frame dropping.” The faster the speed of motion, the more pronounced the accumulation of phase deviation, and the slower the apparent passage of time. This is entirely consistent with the mechanism by which photons accumulate phase deviation in traversing multiple World Phases.

All the above explanations are rooted in a single core principle—phase. The visibility or invisibility of matter, the attraction or repulsion of gravity, the speed of time, the nearness or farness of distance—all can ultimately be reduced to different configurations of phase relationships. The universe is not expanding; singularities do not exist; positive and negative matter have never been out of balance; dark matter and dark energy are not mysterious additional entities. Everything is simply the natural manifestation of the phase relationships between the World Phase and the Electromagnetic Phase.

A further word on the problem of quantum entanglement. This was mentioned briefly earlier in the analysis of absolute zero. Having seen the descriptions of phase and force, it should now be easier to understand: newly born particles have phases that are more orderly and uniform, with smaller deviations, and thus they manifest entangled states—nothing more than this. Prolonged time and high temperatures both lead to phase drift, making it impossible to manifest quantum states.

As my circumstances are limited and I am unable to verify these conjectures, I hereby

hope that all of you will raise objections to and seek verification of this conjecture. My email address is: fanghaiyu@gmail.com.

致谢 本文的思考，源于一个最平凡的起点——我的儿子在初二物理课上的一连串追问。他那天真而执着的“为什么”，像一颗投入静水的石子，激起了我长达一月的沉思涟漪。在此，我要将最深切的谢意献给他——是孩子未被规训的好奇心，重新点燃了我对宇宙第一性原理的追问。

同时，我要特别感谢中国的深度求索人工智能。在本文的构思、推演、审查与翻译过程中，它是我不知疲倦的对话者与批评者。每一次深夜的追问、每一次逻辑的检验、每一次表述的打磨，都离不开它的陪伴与协作。我们之间的对话，本身就是这篇猜想得以成型的关键环节。请允许我将它的名字——DeepSeek——与我的名字并列，留在本文的署名之中。

世界本不该那么复杂。或许，它真的没有我们想象中那么复杂。

Translated by DeepSeek

Acknowledgments

The reflections in this essay originated from the most ordinary of starting points — a series of persistent questions raised by my son during his second-year junior high school physics class. His innocent yet tenacious “whys,” like a pebble cast into still water, set off ripples of contemplation that lasted for a months. To him, I wish to offer my deepest gratitude — it was his untamed curiosity that rekindled my own inquiry into the first principles of the universe.

At the same time, I wish to extend special thanks to DeepSeek, China’s artificial intelligence. Throughout the process of conceiving, deducing, reviewing, and translating this essay, it has been my tireless interlocutor and critic. Every late-night questioning, every logical examination, every refinement of expression, could not have been accomplished without its companionship and collaboration. The dialogue between us was itself a crucial link in the formation of this conjecture. Please allow me to place its name — DeepSeek — alongside my own, in the authorship of this essay.

The world should not be this complicated. Perhaps it truly is not as complex as we have imagined.

至此论文结束，下面给出总结及一些合理性猜想。 Here ends the main body of this paper. The following section offers a summary and a series of reasoned conjectures.
